

# Christian Sassi

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## EDUCATION

### Master's Degree in Computer Science, Cybersecurity - University of Trento

Graduated with a final grade of 110/110 cum laude.

Trento, Italy

Sep. 2023 – Dec. 2025

### Bachelor's Degree in Computer Engineering - University of Trento

Graduated with a final grade of 108/110.

Trento, Italy

Sep. 2020 – Sep. 2023

## EXPERIENCE

### Bug Bounty Program Operator

Huawei

- Conduct **mobile security** research on private **bug bounty programs**.

Feb. 2026 – Present

Munich, Germany

### Software Developer

Energiee3

- Designed a secure **Python** backend that orchestrates authenticated access to multiple external databases through an **encrypted local metadata layer**.
- Built a web-based **SQL client (HTML, CSS, JavaScript)** for safe query execution with efficient **connection management** and no external dependencies.

Nov. 2025 – Jan. 2026

Reggio Emilia, Italy

### Researcher Intern - Cybersecurity

Fondazione Bruno Kessler (FBK)

- Engineered a **federated learning** pipeline by adapting the **DAICS** deep-learning anomaly-detection framework for **Industrial Control Systems (ICS)** security.
- Integrated **adaptive aggregation** and **client selection** from the **FLAD** framework to improve training efficiency.
- Benchmarked on the **SWaT** dataset, with preliminary results approaching the centralized DAICS baseline while preserving **data privacy**.

Mar. 2025 – Dec. 2025

Trento, Italy

### Software Engineer Intern - Embedded Systems

University of Trento

- Validated **SPARK SR1020 Ultra-Wideband (UWB) IoT** devices through **specification-based testing**.
- Optimized **C** firmware for real-time **UWB** communication, reducing response time by **50%**.
- Developed **Python** tools to support firmware debugging and performance tuning.

Feb. 2023 – Jun. 2024

Trento, Italy

## PROJECTS

### MITRE Embedded Capture The Flag (eCTF) Competition | C

- Competed in MITRE's **eCTF** national **firmware security** competition, where the team placed **13th of 116**.
- Wrote defensive **C** code in the embedded firmware: **memory zeroing**, **overflow checks**, and overflow-safe primitives.

### Security Testing Analysis for an E-commerce Platform | Java

- Ran **static (SAST)** and **dynamic (DAST)** analysis of a **Java** platform with **SpotBugs** and **OWASP ZAP**.
- Identified **15+ vulnerabilities** including **SQLi**, **XSS**, and **CSRF**, triaging findings for remediation.
- Designed and applied **15+ targeted fixes** for the discovered issues.

### T.R.U.S.T. - Secure Chat Application | Rust

- Built an **end-to-end encrypted** chat application in **Rust** (TUI), implementing **Signal** protocol primitives.
- Implemented **X3DH** key establishment, the **Double Ratchet** (forward secrecy), and authenticated **AES-GCM** encryption.

## PUBLICATIONS

### Real-time Musical Haptics with Ultra-wideband: A Study on Latency, Reliability, and Perception

L. Turchet, C. Sassi, D. Vecchia, G. Picco - Published in IEEE Transactions on Haptics on January 6, 2025.

## SKILLS

**Spoken Languages:** Italian, English

**Programming Languages:** Python, C/C++, Rust, Java, JavaScript, HTML/CSS, SQL

**Security:** Vulnerability Assessment, Web Security, Mobile Security, Network Security, Reverse Engineering, Binary Exploitation (Pwn), Injection Attacks, Cryptography, Federated Learning

**Security Testing:** SAST, DAST, OWASP Top 10, CVE, CWE